

**PROFESSIONAL EXPERIENCES****National Centre for Infectious Diseases, Singapore***Research Staff (Part-time)***May 2025 - May 2026**

Advisors: Adjunct Assoc. Prof. Kelvin Bryan Tan, Asst. Prof. Jue Tao Lim

**Department of Chemical & Biomolecular Engineering, National University of Singapore, Singapore***Research Engineer (Full-time)***Jul. 2023 - Jan. 2025**

Advisor: Asst. Prof. Zhe Wu

**Department of Biomedical Engineering, National University of Singapore, Singapore***Undergraduate Research Assistant (Internship)***Mar. 2023 - Jul. 2023**

Advisor: Asst. Prof. Yueming Jin

**Rolls-Royce@NTU Corporate Laboratory, Nanyang Technological University, Singapore***Research Engineer (Internship)***May 2022 - Aug. 2022**

Advisors: Dr. Aditya Venkataraman, Dr. Benjamin Cheong

**School of Computing, National University of Singapore, Singapore***Undergraduate Research Assistant (Internship)***Mar. 2022 - May 2022**

Advisor: Prof. Shengdong Zhao

**EDUCATION****Nanyang Technological University, Singapore****Jan. 2025 - present***Ph.D. candidate, Medicine (GPA: 4.88/5.0)*

Advisors: Asst. Prof. Jue Tao Lim, Dr. Ian Wee Liang En (Singapore General Hospital)

**National University of Singapore, Singapore****Aug. 2019 - Jun. 2023***B.Eng. (Hons), Computer Engineering (Distinction, GPA: 4.35/5.0)*

Advisors: Dr. Ganesh Neelakanta Iyer, Dr. Ravi Suppiah, Asst. Prof. Deepu John (University College Dublin)

**AWARDS AND HONORS****Best Oral Presentation**, Resident & Student Category, 20th PHOM Conference**08/2026****Travel Grant**, ESCMID Global 2026**04/2026****Top Rated Abstract**, ESCMID Global 2026**04/2026****NTU Research Scholarship**, NTU**2025 - Present****IEEE Singapore Computer Society Prize**, NUS**06/2023****Outstanding Computing Project Prize**, NUS**06/2023****PUBLICATIONS**

1. **Zihao Wang**, Dariya Nikitin, George T.B. Young, Matan Yechezkel, Liang En Wee, Martin T.W. Chio, Lin Geng, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Joseph A. Lewnard, Lilith K. Whittles, and Jue Tao Lim, "The long-term epidemiological impacts of doxycycline post-exposure prophylaxis and vaccination against multidrug-resistant *Neisseria gonorrhoeae*: a mathematical modelling study," under review by *Science Translational Medicine*. [[medRxiv](#), [GitHub](#)].
2. Lin Geng\*, Perran A. Ross\*, Sui Han Ng\*, Feng Guang Goh, Yu Cai, Tao Huang, Jo Yi Chow, **Zihao Wang**, Esther Li Wen Choo, Chia-Chen Chang, Lisa Couper, Xinyue Gu, Ary Hoffmann, and Jue Tao Lim, "Assessing

the robustness of Wolbachia-mediated incompatible-insect technique to future climate change scenarios,” submitted. [[medRxiv](#)].

3. An Ting Tay\*, Peihong Guo\*, Jinghao Nicholas Ngaim\*, **Zihao Wang\***, Liang En Wee, Calvin Chiew, Lalitha Kurupatham, Lee Ching Ng, Po Ying Chia, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim\*, “Risk of post-acute multi-organ sequelae after dengue reinfection,” submitted.
4. Liang En Wee\*, Reen Wan Li Ho\*, **Zihao Wang\***, Peihong Guo\*, Rhea Khanna, Calvin Chiew, Russell Jingxian Li, Jiahui Li, Chee-Fu Yung, Chia Yin Chong, Zubaidah Said, David Chien Boon Lye, Vernon Lee, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization following paediatric hospitalization for respiratory-syncytial-virus versus influenza or Omicron COVID-19 in a tropical setting: a population-based cohort study,” submitted.
5. Peihong Guo\*, Reen Wan Li Ho\*, **Zihao Wang\***, Ian Liang En Wee\*, Russ Li, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization after hospitalization for COVID-19, influenza or respiratory syncytial virus in adults: a population-based cohort study,” submitted.
6. **Zihao Wang**, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission,” *Nature Health* 1(7), 724-736, 2026. [[Paper](#), [GitHub](#)].
7. **Zihao Wang\***, Yuhan Li\*, Yao Shi, Maher M. Alrashed, Abdulelah S. Alshehri, and Zhe Wu, “Input convex lipschitz recurrent neural networks for robust and efficient process modeling and optimization,” submitted. [[arXiv](#), [GitHub](#)].
8. **Zihao Wang**, and Zhe Wu, “Towards foundation model for chemical reactor modeling: Meta-learning with physics-informed adaptation,” *Chemical Engineering Research and Design* 218, 839-853, 2025. IF: 4.0, JCR Q2. [[Paper](#), [GitHub](#)].
9. **Zihao Wang**, Donghan Yu, and Zhe Wu, “Real-time machine-learning-based optimization using input convex long short-term memory network,” *Applied Energy* 377, 124472, 2025. IF: 12.2, JCR Q1. [[Paper](#), [GitHub](#)].
10. **Zihao Wang**, and Ravi Suppiah, “Upper limb movement recognition utilising EEG and EMG signals for rehabilitative robotics,” *Future of Information and Communication Conference*, 676-695, 2023. [[Paper](#), [GitHub](#)].  
(\* denotes equal contribution)

## PRESENTATIONS

1. **Zihao Wang\***, Dariya Nikitin, George T.B. Young, Matan Yechezkel, Liang En Wee, Martin T.W. Chio, Lin Geng, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Joseph A. Lewnard, Lilith K. Whittles, and Jue Tao Lim, “The long-term epidemiological and evolutionary consequences of joint doxycycline post-exposure prophylaxis and vaccination deployment against multidrug-resistant *Neisseria gonorrhoeae*: a mathematical modelling study,” *20th Singapore Public Health & Occupational Medicine (PHOM) Conference*, Singapore, 2026. (**Oral, Award**).
2. **Zihao Wang\***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission,” *Singapore HIV, Hepatitis and STI Congress (SHHSC)*, Singapore, 2026. (**Oral**).
3. **Zihao Wang\***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Modelling the interplay between doxycycline post-exposure prophylaxis, four-component serogroup B meningococcal (4CMenB) vaccine, and antimicrobial resistance in *Neisseria gonorrhoeae*,” *European Congress of Clinical Microbiology and Infectious Diseases (ESCMID Global 2026)*, Munich, Germany, 2026. (Poster, **Top 2%**).
4. **Zihao Wang\***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Modelling the interplay between doxycycline post-exposure prophylaxis, four-component meningococcal serogroup B (4CMenB) vaccine, and antimicrobial resistance in *Neisseria gonorrhoeae*,” *LKCMedicine EARly Researcher Network (LEARN) Symposium*, Singapore, 2026. (**Flash Talk**).

5. **Zihao Wang\***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-run public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England,” *Epidemics: 10th International Conference on Infectious Disease Dynamics (Epidemics10)*, San Diego, California, USA, 2025. **(Oral)**.
6. **Zihao Wang\***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England,” *Singapore Health & Biomedical Congress (SHBC)*, Singapore, 2025. (Poster).
7. **Zihao Wang**, Wenlong Wang\*, and Zhe Wu, “Accelerated modeling of various chemical processes using meta-learning-based foundation models: A few-shot learning approach using Reptile,” *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. **(Oral)**. [[Abstract](#)].
8. **Zihao Wang**, Wenlong Wang\*, and Zhe Wu, “Input convex LSTM for fast machine learning-based optimization,” *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (Poster). [[Abstract](#)].
9. **Zihao Wang**, and Zhe Wu\*, “Input convex LSTM: A convex approach for fast model predictive control,” Institute for Machine Learning (Head: Professor Sepp Hochreiter), Johannes Kepler University, Linz, Austria, December 2023. **(Invited talk by Professor Günter Klambauer)**.  
(\* denotes presenting author)

## DISSERTATIONS

1. **Zihao Wang**, “Multimodal sensor fusion using AI for healthcare IoT sensors,” B.Eng. Dissertation, Computer Engineering, National University of Singapore, Singapore, 2023. [[Paper](#)].
2. **Zihao Wang**, “AI research for rehabilitative robotics,” Undergraduate Research Opportunity Project (UROP) Thesis, Computer Engineering, National University of Singapore, Singapore, 2022. [[Paper](#)].

## SKILLS

Computer: C, C++, Java, Python, R, Assembly Language, MATLAB, PostgreSQL, Verilog, Vensim, Gephi, LaTeX  
 Languages: Mandarin (native), English (bilingual), Cantonese (proficient), Korean (conversational)